A new (a, b)-class atom $h_{(7,2)}(q)$ and the first multi-shape σ_1-coincidence at $n = 8$

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Abstract

For each partition λ of n let $\sigma_1(V_\lambda)(q)$ denote the trace of the staircase Hecke product $\Pi_q^{(n)} = (1+T_1)(1+T_2)(1+T_1)(1+T_3)(1+T_2)(1+T_1)\cdots(1+T_{n-1})\cdots(1+T_1)$ acting on the Specht module V_λ via Young's seminormal form. Empirically

$$\sigma_1(V_\lambda)(q) \;=\; q^{a_\lambda}(1+q)^{b_\lambda}\, P_\lambda(q),$$

with P_λ palindromic and (a_λ, b_λ) maximal among factorizations of this shape. Define the (a, b)-*class* of λ as the equivalence class of partitions sharing (a_λ, b_λ).

We exhibit the first multi-element (a, b)-class: $\lambda = (3, 3, 2)$ and $\mu = (4, 2, 1, 1)$ are both partitions of 8, both self-conjugate, both have $(a, b) = (7, 2)$, and they have the *identical* σ_1 polynomial despite $\dim V_\lambda = 42 \neq 90 = \dim V_\mu$. The common factor $h_{(7,2)}(q)$ is irreducible palindromic of degree 12 with leading coefficient 1; it is novel — not in any previously catalogued atom-alphabet position. We rule out W-graph local structure as an explanation by direct comparison of τ-multisets, and reframe the coincidence as a Markov-trace cancellation $\sum_\nu |C_\nu|\, P_\nu(q)\, \delta(c_\nu) \equiv 0$ in $\mathbb{Q}[q]$, with $\delta = \chi_\lambda - \chi_\mu$ supported on 8 of 22 conjugacy classes including the identity, where δ has defect $-48 = \dim V_\lambda - \dim V_\mu$.

1 Introduction

Let $\mathcal{H}_q(S_n)$ denote the Iwahori–Hecke algebra of type A_{n-1} over $\mathbb{Q}(q)$, with generators $T_1, \ldots, T_{n-1}$ and quadratic relation $T_i^2 = (q-1)T_i + q$. Fix the staircase reduced word

$$\mathbf{w}_0 = (s_1)(s_2 s_1)(s_3 s_2 s_1)\cdots(s_{n-1}\cdots s_1)$$

for the longest element $w_0 \in S_n$. Define the staircase Hecke product

$$\Pi_q^{(n)} \;=\; \prod_{i=1}^{n-1}\prod_{j=i}^{1}(1+T_j) \;=\; (1+T_1)(1+T_2)(1+T_1)(1+T_3)(1+T_2)(1+T_1)\cdots$$

on $\binom{n}{2}$ factors. For each partition $\lambda \vdash n$ let V_λ be the Specht module realized via Young's seminormal form, and define the σ_1-trace

$$\sigma_1(V_\lambda)(q) \;=\; \mathrm{tr}\big(\Pi_q^{(n)} \,\big|\, V_\lambda\big) \;\in\; \mathbb{Z}[q].$$

Definition 1 ((a, b)-class)**.** The unique factorization $\sigma_1(V_\lambda)(q) = q^{a_\lambda}(1+q)^{b_\lambda}P_\lambda(q)$ with $P_\lambda(0) \neq 0$, $P_\lambda(-1) \neq 0$, and P_λ palindromic, defines the pair $(a_\lambda, b_\lambda) \in \mathbb{N}^2$. We call this pair the (a, b)-*degree* of λ, and the equivalence relation $\lambda \sim \mu \iff (a_\lambda, b_\lambda) = (a_\mu, b_\mu)$ partitions $\{\lambda \vdash n\}$ into (a, b)-*classes*.

We exclude rank-isolated shapes ($\ell(\lambda) > (n+1)/2$, where Π_q acts as zero on V_λ) from the enumeration.

Observation 2 (Main observation)**.** At $n = 7$, every (a, b)-class is a singleton: there are 11 non-zero classes, all of size 1. At $n = 8$, there are 14 non-zero (a, b)-classes, exactly one of which has size 2:

$$\big\{(3,3,2),\ (4,2,1,1)\big\}, \qquad (a,b) = (7,2).$$

Moreover $\sigma_1(V_{(3,3,2)})(q) = \sigma_1(V_{(4,2,1,1)})(q)$ as polynomials in q, despite $\dim V_{(3,3,2)} = 42$ and $\dim V_{(4,2,1,1)} = 90$. The common polynomial $P(q)$ is irreducible palindromic of degree 12 over $\mathbb{Q}$.

The remainder of this note documents and partially structures this finding.

2 The Rosetta Stone factorization

The factorization $\sigma_1(V_\lambda) = q^{a_\lambda}(1+q)^{b_\lambda} P_\lambda(q)$ is the per-irrep instance of what we have been calling the *Rosetta Stone*: the trace of $\Pi_q^{(n)}$ on each Specht module splits cleanly into

- a q-power q^{a_λ} (Lusztig a-value, content shift),
- a $(1+q)^{b_\lambda}$ factor (Soergel / categorical multiplicity), and
- a palindromic remainder $P_\lambda(q)$ (the *atom*).

This per-irrep factorization was first verified explicitly at $n = 4$ on April 22, 2026 (cf. memory entry `rosetta_stone_n4_verified`); the present work confirms it through $n = 8$ and isolates the new degree-12 atom $h_{(7,2)}(q)$ described below.

For all $\lambda \vdash 8$ that are not rank-isolated, $P_\lambda(q)$ is palindromic in our computations; this is consistent with the W-graph parity-sheaf self-duality predicted by the monodromic Hecke hypothesis.

3 (a, b)-class enumeration at $n = 7$ and $n = 8$

We computed $\sigma_1(V_\lambda)(q)$ for every non-rank-isolated $\lambda \vdash 7, 8$ via Lagrange interpolation from 32 exact $\mathbb{Q}$-evaluations of the seminormal-form action of $\Pi_q^{(n)}$ on V_λ. Table 1 reports the (a, b)-degrees for $n = 7$; Table 2 for $n = 8$. Every P_λ in both tables is palindromic.

4 The polynomial $h_{(7,2)}(q)$

Theorem 3 (Common atom)**.** $\sigma_1(V_{(3,3,2)})(q) = \sigma_1(V_{(4,2,1,1)})(q) = q^7(1+q)^2\, h_{(7,2)}(q)$, *where*

$$\begin{aligned} h_{(7,2)}(q) \ &= 1 + 27q + 325q^2 + 2169q^3 + 8428q^4 + 19143q^5 \\ &\quad + 25218q^6 \\ &\quad + 19143q^7 + 8428q^8 + 2169q^9 + 325q^{10} + 27q^{11} + q^{12}. \end{aligned}$$

The polynomial $h_{(7,2)}$ is palindromic of degree 12, has leading and constant coefficient 1, and is irreducible over $\mathbb{Q}$.

λ	$\dim V_\lambda$	a	b	$\deg P$
$(2,2,2,1)$	14	9	3	0
$(3,2,1,1)$	35	7	1	6
$(3,2,2)$	21	6	1	8
$(3,3,1)$	21	5	1	10
$(4,1,1,1)$	20	6	3	6
$(4,2,1)$	35	4	1	12
$(4,3)$	14	3	3	12
$(5,1,1)$	15	3	5	10
$(5,2)$	14	2	5	12
$(6,1)$	6	1	11	8
(7)	1	0	21	0

Table 1: (a,b)-degrees at $n=7$, non-rank-isolated shapes. All 11 classes are singletons.

Computational verification. Both σ_1-polynomials are computed by exact Lagrange interpolation in $\mathbb{Q}[q]$ from 32 evaluations of $\mathrm{tr}(\Pi_q^{(8)}|V_\lambda)$ via Young's seminormal form (matrix size 42×42 for $\lambda=(3,3,2)$ and 90×90 for $\lambda=(4,2,1,1)$); both interpolations return the same 22-coefficient polynomial $q^7(1+q)^2h(q)$ with the coefficients displayed above.

Palindromy: the coefficient list $(1,27,325,2169,8428,19143,25218,19143,8428,2169,325,27,1)$ is symmetric, so $q^{12}h_{(7,2)}(1/q)=h_{(7,2)}(q)$.

Irreducibility: a SymPy factorization confirms that $h_{(7,2)}(q)$ is irreducible over $\mathbb{Q}$. No cyclotomic polynomial divides it (a quick check evaluates $h_{(7,2)}$ at all primitive roots of unity of order ≤ 13). No previously catalogued atom of degree ≤ 12 from the $n\leq 7$ alphabet (see the memory entry `atom_alphabet`) is a factor. □

The values

$$h_{(7,2)}(0)=1,\qquad h_{(7,2)}(1)=85404=2^2\cdot 3\cdot 11\cdot 647,$$

are noteworthy: the prime factor 647 appears in no previously catalogued atom value $h(1)$, and the central coefficient $25218=2\cdot 3^2\cdot 1401$ does not match any Eulerian, Narayana, or derangement-RTL value at the relevant length. We therefore record $h_{(7,2)}(q)$ as a *new entry* in the atom alphabet, distinct from every entry isolated through $n=7$.

The σ_1 value at $q=1$ is

$$\sigma_1(V_{(3,3,2)})(1)=\sigma_1(V_{(4,2,1,1)})(1)=1^7\cdot 2^2\cdot 85404=341616=2^4\cdot 3\cdot 11\cdot 647.$$

5 Self-conjugacy and Lusztig a-value

Both $\lambda=(3,3,2)$ and $\mu=(4,2,1,1)$ are self-conjugate ($\lambda=\lambda'$, $\mu=\mu'$). Recall that $n(\lambda)=\sum_{i\geq 1}(i-1)\lambda_i$ is Lusztig's a-invariant for type A. Direct computation gives

$$n(\lambda)=0\cdot 3+1\cdot 3+2\cdot 2=7,\qquad n(\mu)=0\cdot 4+1\cdot 2+2\cdot 1+3\cdot 1=7,$$

and by self-conjugacy $n(\lambda)=n(\lambda')$, $n(\mu)=n(\mu')$, so λ and μ have the *same Lusztig a-value*, namely $a(\lambda)=a(\mu)=7$. This matches our $a_\lambda=a_\mu=7$ in the (a,b)-degree. In particular the q-power exponent in the Rosetta Stone factorization equals the Lusztig a-value here. Both shapes therefore have the same Lusztig two-sided cell-degree contribution; this is a necessary condition for the σ_1-coincidence but, as the next section shows, far from sufficient.

λ	$\dim V_\lambda$	a	b	$\deg P$
$(2,2,2,2)$	14	12	4	0
$(3,2,2,1)$	70	9	2	8
$(3,3,1,1)$	56	8	2	10
$\mathbf{(3,3,2)}$	**42**	**7**	**2**	**12**
$\mathbf{(4,2,1,1)}$	**90**	**7**	**2**	**12**
$(4,2,2)$	56	6	2	14
$(4,3,1)$	70	5	2	16
$(4,4)$	14	4	4	16
$(5,1,1,1)$	35	6	4	12
$(5,2,1)$	64	4	2	18
$(5,3)$	28	3	4	18
$(6,1,1)$	21	3	8	14
$(6,2)$	20	2	8	16
$(7,1)$	7	1	16	10
(8)	1	0	28	0

Table 2: (a,b)-degrees at $n=8$, non-rank-isolated shapes. The unique multi-element (a,b)-class $\{(3,3,2),(4,2,1,1)\}$ is bolded.

6 W-graph local structure does not explain the coincidence

For each standard Young tableau T of shape $\lambda \vdash n$ let $\mathrm{Des}(T) = \{i \in \{1,\dots,n-1\} : i+1 \text{ lies strictly below } i \text{ in } T\}$ be its descent set. The multiset of descent sets $\{\mathrm{Des}(T) : T \in \mathrm{SYT}(\lambda)\}$ records the local W-graph structure attached to the Kazhdan–Lusztig left cell of shape λ.

Proposition 4. *The descent multisets of* $(3,3,2)$ *and* $(4,2,1,1)$ *differ strongly. Even after dimension-normalization:*

- $\lambda = (3,3,2)$*: descent-set sizes are* $\{2,3,4,5\}$ *with normalized weights* $(\frac{1}{14},\frac{3}{7},\frac{3}{7},\frac{1}{14})$*;*
- $\mu = (4,2,1,1)$*: descent-set sizes are* $\{3,4\}$ *with normalized weights* $(\frac{1}{2},\frac{1}{2})$*.*

In particular λ *has* 3 *tableaux with descent set of size* 5*;* μ *has none.* μ *has* 33 *tableaux with descent set of size* 3 *that do not appear among the descent sets of* λ*. The intersection of the descent-set supports has* 33 *elements out of* $|\lambda| = 35$ *and* $|\mu| = 75$*.*

Proof. Direct enumeration over $\mathrm{SYT}((3,3,2))$ (42 tableaux) and $\mathrm{SYT}((4,2,1,1))$ (90 tableaux); see the script `tau_multisets.py` in the supporting computation directory. □

Corollary 5. *The* σ_1*-coincidence is not visible at the W-graph descent level. In particular it is not a shadow of any local descent-multiset agreement, and therefore is not predicted by any parity-sheaf or local KL-graph isomorphism between the two cells.*

7 Markov-trace reframing

Decompose $\Pi_q^{(n)} \in \mathcal{H}_q(S_n)$ in the T_w basis, $\Pi_q^{(n)} = \sum_{w\in S_n} c_w(q)\, T_w$, and let $\widetilde{P}_\nu(q) = \sum_{w\in C_\nu} c_w(q)$ where $C_\nu \subset S_n$ is the conjugacy class of cycle type ν. Each Hecke character χ_λ^q evaluates linearly on T_w; projecting to ordinary characters at $q \to 1$ yields

$$\sigma_1(V_\lambda)(q) \;=\; \sum_{\nu\vdash n} P_\nu(q)\, \chi_\lambda(c_\nu),$$

for an appropriate set of class polynomials $P_\nu(q)$ adapted to the Geck–Pfeiffer character formula on the Hecke algebra. In particular

$$\sigma_1(V_\lambda)(q) - \sigma_1(V_\mu)(q) \;=\; \sum_{\nu \vdash n} P_\nu(q)\,\delta(c_\nu), \qquad \delta = \chi_\lambda - \chi_\mu.$$

Proposition 6 (Support of δ at $n = 8$)**.** *For $\lambda = (3,3,2)$ and $\mu = (4,2,1,1)$, the difference $\delta = \chi_\lambda - \chi_\mu$ is supported on exactly 8 of the 22 conjugacy classes of S_8:*

ν	$\chi_\lambda(c_\nu)$	$\chi_\mu(c_\nu)$	$\delta(c_\nu)$
$(7,1)$	0	-1	1
$(5,3)$	-1	0	-1
$(5,1,1,1)$	2	0	2
$(4,2,1,1)$	-2	2	-4
$(3,2,2,1)$	2	0	2
$(3,1,1,1,1,1)$	-6	0	-6
$(2,2,1,1,1,1)$	2	-6	8
(1^8)	42	90	-48

On the other 14 conjugacy classes both characters agree.

Proof. Murnaghan–Nakayama recursion via β-numbers, performed in exact integer arithmetic, with the sanity check $\sum_\lambda \dim(V_\lambda)^2 = 8! = 40320$ passing; full character table on the two rows is recorded in `2026-04-30-tcore-test/RESULT.md`. □

Corollary 7. *The σ_1-coincidence $\sigma_1(V_{(3,3,2)})(q) = \sigma_1(V_{(4,2,1,1)})(q)$ in $\mathbb{Z}[q]$ is equivalent to the Markov-trace identity*

$$\sum_{\nu \in \mathrm{supp}(\delta)} P_\nu(q)\,\delta(c_\nu) \;\equiv\; 0 \quad \text{in } \mathbb{Q}[q].$$

At $q = 1$, this becomes a non-trivial linear identity among the eight values $\{1, -1, 2, -4, 2, -6, 8, -48\}$ weighted by the $q = 1$ values of the eight class polynomials.

Remark. The defect at the identity, $\delta(c_{(1^8)}) = -48 = 42 - 90$, must be exactly cancelled by contributions from the other seven classes in $\mathrm{supp}(\delta)$. Three of those classes — $(4,2,1,1)$, $(2,2,1,1,1,1)$, and (1^8) — have both χ_λ and χ_μ nonzero, so the identity is genuinely q-weighted: it cannot be reduced to an obvious character-zero mechanism (such as a t-core / Type-II coincidence).

8 Conjecture: every multi-element (a, b)-class carries an atom

Conjecture 8 (Class atom)**.** For every n and every multi-element (a,b)-class $[\lambda] \subset \{\nu \vdash n : (a_\nu, b_\nu) = (a,b)\}$, the greatest common divisor

$$h_{[\lambda]}(q) \;:=\; \gcd_{\mu \in [\lambda]} P_\mu(q)$$

is a non-trivial palindromic polynomial in $\mathbb{Z}[q]$, the (a,b)-*class atom*. Moreover $h_{[\lambda]}$ enters the atom alphabet as a new entry that is not shared with any singleton class P_ν at the same n.

The polynomial $h_{(7,2)}(q)$ above is the first instance of a class atom: the (a,b)-class $\{(3,3,2),(4,2,1,1)\}$ at $n = 8$ has $P_{(3,3,2)} = P_{(4,2,1,1)} = h_{(7,2)}$, so the gcd is the whole common P. At $n \le 7$ no multi-element class exists, so the conjecture has no content there.

9 Open questions

1. **Hecke class polynomials.** Compute the Geck–Pfeiffer class polynomials $P_\nu(q)$ for $\Pi_q^{(8)}$ explicitly, and verify the cancellation identity $\sum_\nu |C_\nu|\, P_\nu(q)\, \delta(c_\nu) \equiv 0$ in $\mathbb{Q}[q]$ directly. This requires expanding $\Pi_q^{(8)}$ in the T_w basis (28-letter staircase word, of order 10^6 terms in $\mathbb{Z}[q]$); feasible in plain Python.

2. **Categorical home of $h_{(7,2)}$.** The atom alphabet isolated through $n = 7$ has explained entries (Eulerian E_2, Narayana, derangement-RTL, A_9, A_{10}, the orphan A_{11} from `novel_atom_deg10`). Where does $h_{(7,2)}$ live? One candidate is a singular-Soergel atomic Leibniz contribution in the Patimo categorical framework: the $(a, b) = (7, 2)$ cell S_8 admits two cells of the same Lusztig a-value but distinct two-sided structure, which could produce a single Hochschild–type contribution shared between them.

3. **Higher n.** At $n = 9, 10$, how many multi-element (a, b)-classes appear, and do they all have non-trivial class atoms? Is the multiplicity of multi-element classes monotone in n? Are class-atom polynomials closed under $n \mapsto n + 1$ branching, or do they form an independent layer of atoms?

4. **Self-conjugacy.** Both members of the $(7, 2)$ class at $n = 8$ are self-conjugate. Is this generic, i.e. do all multi-element (a, b)-classes consist entirely of self-conjugate shapes? If so, this restricts the conjectural class-atom alphabet to a controlled subset of irreducible palindromes.

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Computational data. Source code and exact polynomial values are recorded under `/home/clio/projects/scra` and `/home/clio/projects/scratch/2026-04-30-sigma1-equality-structure/`; the Murnaghan–Nakayama character verification of the δ-support is recorded under `/home/clio/projects/scratch/2026-04-30-t`